

Vivek L. Kale

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Github: github.com/vlkale | Website: vlkale.github.io | U.S. citizen with Secret clearance

Professional Summary

- Software engineer and technical leader specializing in performance optimization and reliability of scientific and AI applications deployed on GPU-based clusters.
- Proven expertise in developing systems software to optimize CPU and GPU computation and multi-GPU communication in applications; demonstrated leadership in managing technical teams and driving open-source initiatives for delivering production-grade runtimes, profiling+debugging tools and compiler support for GPU-based parallel programming.
- Strong track record of technical management and cross-functional collaboration for production software, multi-million dollar research initiatives, peer-reviewed publications, proposals for programming standards, and partnerships with hardware vendors and research institutions.

Education

- **Certification:** Technical Management Program, 2024, University of California at Los Angeles
- **Ph.D.** (Doctor of Philosophy), Computer Science, 2015, University of Illinois at Urbana-Champaign
 - Specialization: *Parallel Computing and HPC*
 - Thesis topic: *Low-overhead OpenMP Loop Scheduling to Scale MPI Communication*
- **B.S.** (Bachelor of Science), Computer Science, 2007, University of Illinois at Urbana-Champaign

Relevant Experience

Technical Leadership and Industry-grade Open-source HPC Software

Sandia National Laboratories

Principal Member of Technical Staff II

July 2024 – Present

- Led cross-functional collaboration within Sandia/DOE to technically manage and deploy Kokkos Tools on GPU-based clusters, considering key integration points, e.g., LLVM, MLIR, JIT, Kokkos Kernels, Kokkos-LLM, code translation, PyTorch, visualization dashboards, benchmarking, and AI4Science workflows.
- Built a team from 1 to 8 to develop tools for Kokkos and C++ StdPar, i.e., base-language parallelism.
- Built awareness for Kokkos Tools via a blog post (kokkos.org), a keynote (ISCC25) and talks, e.g., HPSF25.
- Designed and implemented software for tools for distributed GPU programming with Kokkos, specifically (1) inter-process profiling and adaptivity via PMPI and (2) job-level monitoring and feedback via LDMS.
- Augmented Kokkos Tools runtime auto-tuning via low-level metrics from, e.g., PAPI, CUPTI, instead of timings, reducing time to converge to optimal Kokkos tuning parameters by 32%.
- Released a Spack package for Kokkos Tools, resulting in 16 new users of Kokkos Tools.
- Developed AI-assisted HPC Tools through LLMs (coderoSETTA.com) and autotuning (TAU+APEX) for Kokkos applications run on NVIDIA GPUs, resulting in a poster presentation at GTC 2025.
- Leading industry partnerships on pathfinding for application deployment with Kokkos Tools on NeoClouds, e.g., CoreWeave Cloud, and on dataflow accelerators, e.g., Cerebras WSE-3, Tensorrent Wormhole.
- Developing energy-aware optimization for Kokkos via NVML, achieving 4x energy reduction.
- Implementing methods for verification and validation of Kokkos Tools tooling data by associating it with formal methods and performance models.

Senior Member of Technical Staff

August 2022 – July 2024

- Developed and maintained Kokkos Tools for the CMake build system, low tooling overheads, CI/CD, auto-tuning, and vendor tools integration, leading to 15 merged github PRs.
- Developed a debugging tool that detected 7 common Kokkos user bugs by analyzing LLVM IR of Kokkos programs via symbolic execution, leading to a paper at SC24's Correctness workshop.
- Implemented 5 new loop transformation features in LLVM OpenMP, leading to a 1.7x speedup for an OpenMP+CUDA benchmark using the index set split feature and 3 standardized features in OpenMP 6.0.
- Implemented OpenMP multi-GPU parallelism features in a prototype library for LLVM, leading to a 1.3x speedup over a CUDA-aware MPI approach, and 9 multi-GPU feature proposals for OpenMP 7.0.
- Led pathfinding and partnerships with major CSPs (AWS, Google Cloud, Microsoft Azure, OCI) on tooling for the deployment of DOE science simulations on cloud, exascale+cloud and multi-cloud platform setups.

Brookhaven National Laboratory Assistant Computational Scientist

May 2019 – August 2022

- Implemented OpenMP user-defined multi-GPU scheduling for LLVM, offering 2.1x speedup over using MPI parallelization, leading to papers at IWOMP 2020 and BCB 2021.
- Implemented performance optimizations in LLVM for OpenMP asynchronous GPU offloading that achieved a 1.2x speedup, leading to a paper at SC22's HiPar workshop.
- Developed performance benchmarks that evaluated 5 major vendor OpenMP GPU implementations, leading to an ACM journal paper and an IWOMP 2021 workshop paper.
- Demonstrated technical leadership as technical project manager for the ECP SOLLVE project, submitting 12 ECP milestone reports, organizing 7 GPU hackathons, and defining 3 project KPIs.

HPC Software Development and Performance Engineering

USC/ISI + Charmworks, Inc.

Software Engineer

December 2015 – May 2019

- Implemented User-defined Loop Schedules (UDS) for OpenMP and RAJA via a prototype library for LLVM and GCC, leading to a paper at IWOMP 2018 and 3 github PRs merged in Charm++.
- Performance analysis and optimization of MPI+CUDA scientific applications on NVIDIA GPUs via CUPTI and auto-tuning, leading to 1.4x speedup of an application for computer chip design.
- Developed novel and efficient multi-level loop schedulers in Charm++, leading to a 1.2x speedup on the PRK particle-in-cell benchmark code and a Best Poster Candidate at SC18.

LLNL + UIUC

Researcher

Jun 2010 – Dec 2015

- Implemented a ROSE-based compiler pass and PMPI-based runtime system for MPI+OpenMP applications to use loop scheduling techniques, leading to a 1.4x speedup on a multicore cluster.
- Implemented shared memory extensions for MPICH, leading to a paper with 140+ citations.
- Implemented multicore and GPU performance optimizations for domains of linear algebra, blood flow, fusion, and combustion, leading to 2 papers at IPDPS.

General Software Development

Proteus Technologies + Wolfram

Software Developer

August 2007 – September 2008

- Developed and tested service-oriented software to monitor the health of a large-scale distributed system for the US government, leading to an internal white paper and software package.
- Implemented functionality in Mathematica for users to send emails from within a Mathematica evaluation kernel, via sendmail and TLS, leading to a new software feature in Mathematica.

Technical Skills

- **Languages:** C, C++, Python, Fortran, Java, bash, csh, VHDL, Matlab
- **Software:** OpenMP and OpenACC (GCC, LLVM), CUDA, HIP, Kokkos, C++ StdPar, MPI (MPICH), NCCL, MLIR, Kokkos Kernels, LAMMPS, PyTorch, numpy, pandas, scikitlearn, docker, slurm, Kubernetes
- **Tools:** Kokkos Tools, PMPI, ompt, PAPI, nvtx, roctx, NVIDIA Nsight Systems+Compute, HPCToolkit, CUPTI, tau, VTune, clang-tidy, clang-format, KLEE, gprof, nvprof, gdb, Spack, CMake
- **Utilities:** Cursor AI, Claude Code, git, vi, gnuplot, emacs, autoconf, LaTeX, matplotlib

Open-source Software Projects

1. **OpenMP Locality-aware Loop Scheduling:** <https://github.com/vlkale/lw-sched>
2. **Multi-GPU Stencil Benchmark:** <https://github.com/vlkale/ParallelProgrammingWithOpenACC>
3. **MPI Slack Predictor:** <https://github.com/vlkale/slack-trace>